

Park 2016 Table-4 OR vs mean |SHAP| rank (migraine/park)

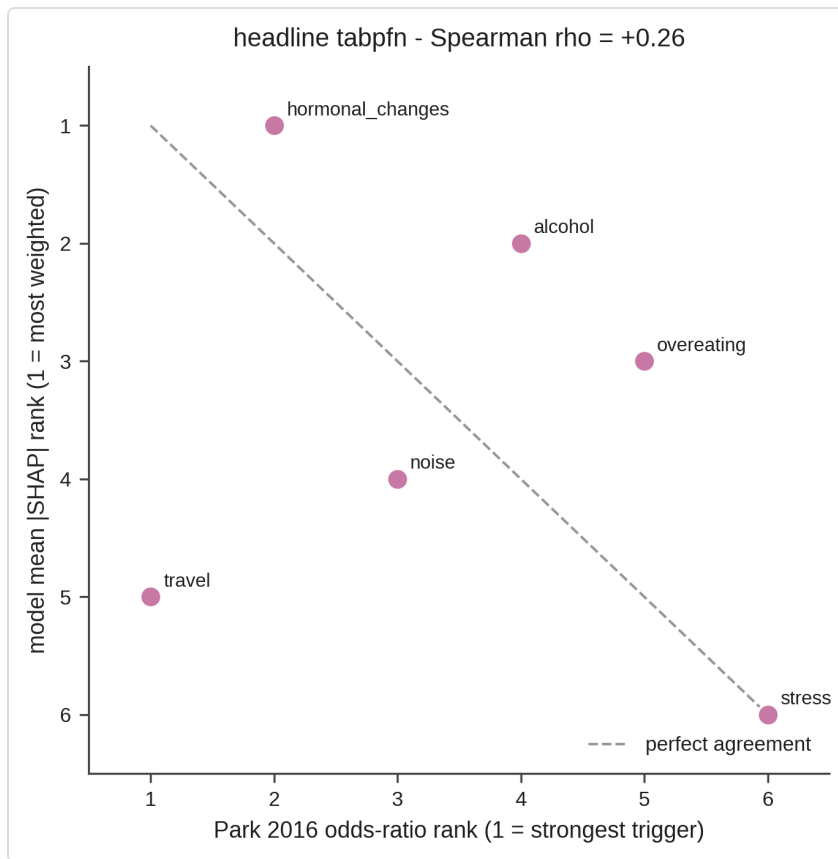
What this compares. Park et al. 2016 report odds ratios for self-reported migraine triggers in the SHD cohort - a population-level, same-day association between a trigger and a migraine day. The model's mean |SHAP| rank is a data-driven importance for *next-day* individual prediction from the park trigger features. The two answer different questions, so this is a convergence check, not a validation: a positive Spearman correlation means the model recovers Park's trigger ordering; a value near zero or negative means it weights the triggers differently, which is expected when same-day cross-sectional odds ratios are asked to drive a next-day forecast on a small feature set. Each panel shows the leaf's full provenance (architecture, hyperparameter-tuning, data-split ratio and split type) and the odds-ratio-rank vs SHAP-rank scatter against the agreement diagonal.

headline: tabpfn (tabpfn, mean |SHAP|)

tabpfn/version_3-binary [NonHP (library defaults)] | split 70_30/chrono | hold-out AUROC 0.646, AUPRC 0.126, calib +1.05

Spearman correlation (SHAP rank vs Park OR rank): **+0.257** over 6 triggers.

trigger	Park OR	OR rank	SHAP rank	mean SHAP
travel_today	6.4	1	5	0.0153
hormonal_changes_today	3.5	2	1	0.0532
noise_today	2.8	3	4	0.0194
alcohol_today	2.5	4	2	0.0228
overeating_today	2.4	5	3	0.0219
stress_today	1.8	6	6	0.0151

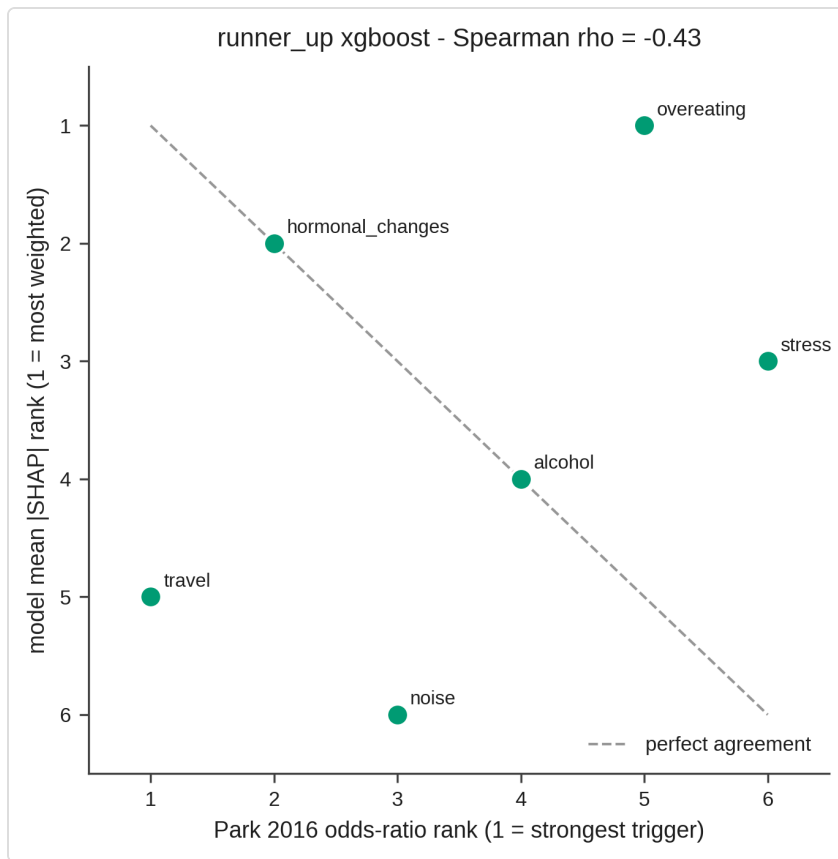


runner_up: stacked_2xgb_meta_lr (xgboost, mean |SHAP|)

stacked_2xgb_meta_lr [NonHP (library defaults)] | split 70_30/chrono | hold-out AUROC 0.625, AUPRC 0.115, calib +0.82

Spearman correlation (SHAP rank vs Park OR rank): **-0.429** over 6 triggers.

trigger	Park OR	OR rank	SHAP rank	mean SHAP
travel_today	6.4	1	5	0.0088
hormonal_changes_today	3.5	2	2	0.0279
noise_today	2.8	3	6	0.0054
alcohol_today	2.5	4	4	0.0157
overeating_today	2.4	5	1	0.0292
stress_today	1.8	6	3	0.0205

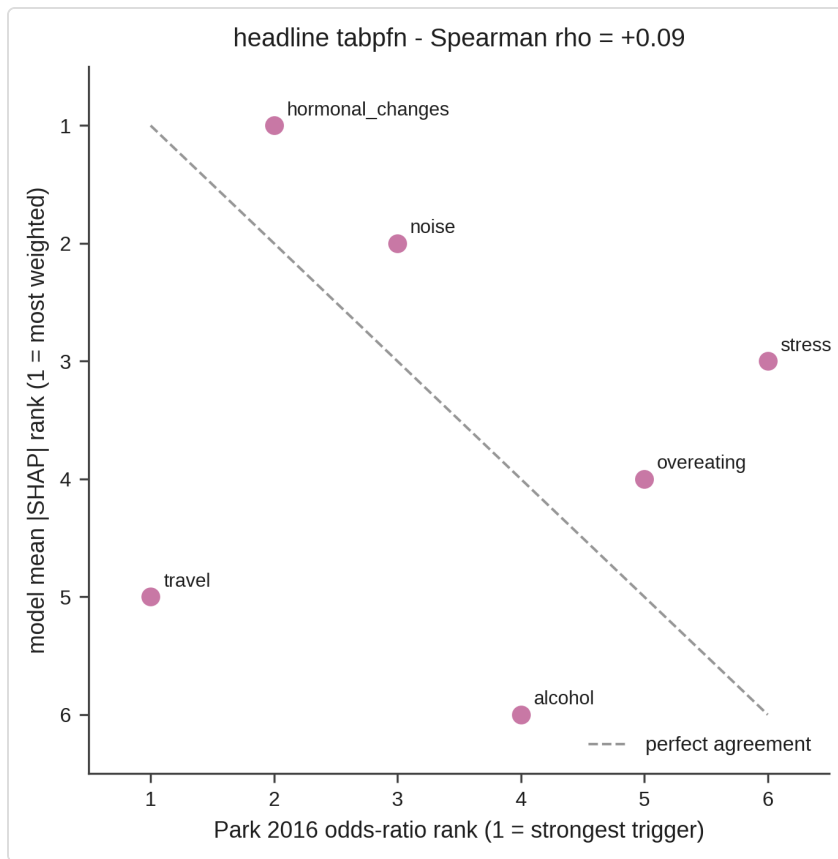


headline: tabpfn (tabpfn, mean |SHAP|)

tabpfn/version_3-default [NonHP (library defaults)] | split 80_20/stratified | hold-out AUROC 0.618, AUPRC 0.119, calib +0.59

Spearman correlation (SHAP rank vs Park OR rank): **+0.086** over 6 triggers.

trigger	Park OR	OR rank	SHAP rank	mean SHAP
travel_today	6.4	1	5	0.0045
hormonal_changes_today	3.5	2	1	0.0136
noise_today	2.8	3	2	0.0099
alcohol_today	2.5	4	6	0.0034
overeating_today	2.4	5	4	0.0057
stress_today	1.8	6	3	0.0075

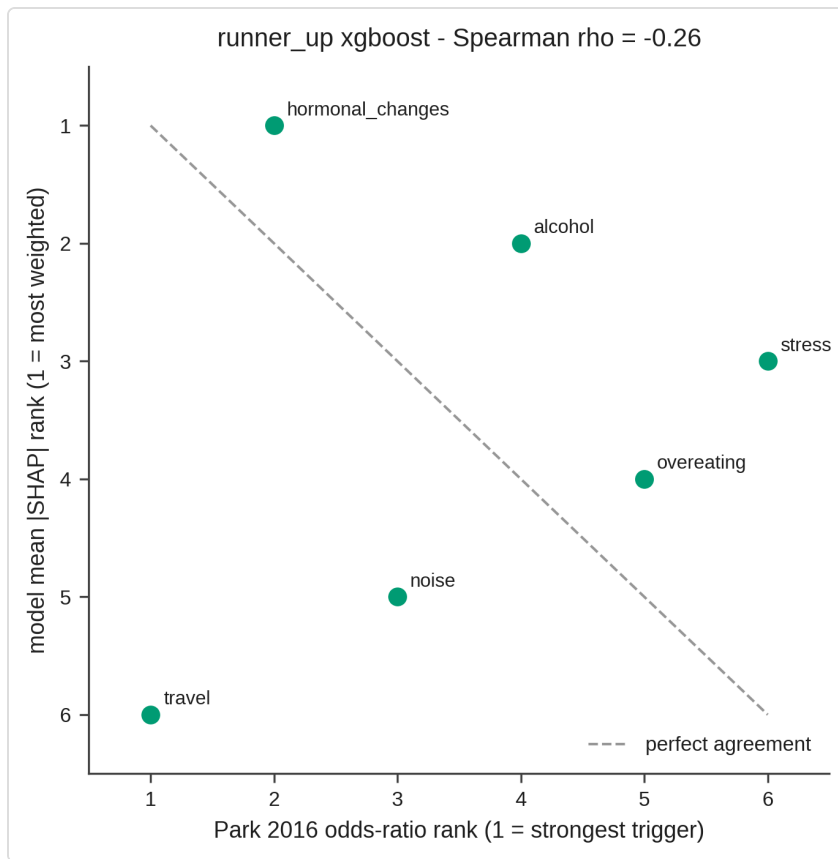


runner_up: stacked_2xgb_meta_lr (xgboost, mean |SHAP|)

stacked_2xgb_meta_lr [NonHP (library defaults)] | split 70_30/stratified | hold-out AUROC 0.576, AUPRC 0.117, calib +0.56

Spearman correlation (SHAP rank vs Park OR rank): **-0.257** over 6 triggers.

trigger	Park OR	OR rank	SHAP rank	mean SHAP
travel_today	6.4	1	6	0.0012
hormonal_changes_today	3.5	2	1	0.0338
noise_today	2.8	3	5	0.0023
alcohol_today	2.5	4	2	0.0115
overeating_today	2.4	5	4	0.0043
stress_today	1.8	6	3	0.0095

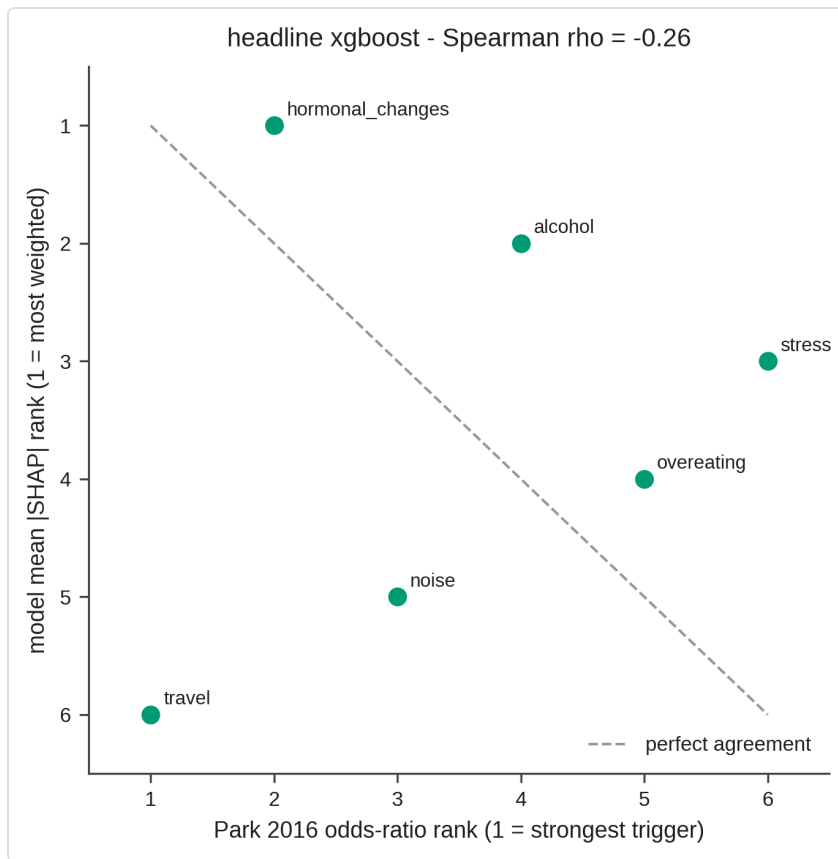


headline: stacked_2xgb_meta_lr (xgboost, mean |SHAP|)

stacked_2xgb_meta_lr [NonHP (library defaults)] | split 70_30/patient | hold-out AUROC 0.541, AUPRC 0.086, calib +0.67

Spearman correlation (SHAP rank vs Park OR rank): **-0.257** over 6 triggers.

trigger	Park OR	OR rank	SHAP rank	mean SHAP
travel_today	6.4	1	6	0.0000
hormonal_changes_today	3.5	2	1	0.0001
noise_today	2.8	3	5	0.0000
alcohol_today	2.5	4	2	0.0001
overeating_today	2.4	5	4	0.0000
stress_today	1.8	6	3	0.0000



runner_up: tabpfn (tabpfn, mean |SHAP|)

tabpfn/version_3-default [NonHP (library defaults)] | split 70_30/patient | hold-out AUROC 0.536, AUPRC 0.082, calib +0.18

Spearman correlation (SHAP rank vs Park OR rank): **-0.029** over 6 triggers.

trigger	Park OR	OR rank	SHAP rank	mean SHAP
travel_today	6.4	1	6	0.0055
hormonal_changes_today	3.5	2	1	0.0220
noise_today	2.8	3	4	0.0081
alcohol_today	2.5	4	2	0.0087
overeating_today	2.4	5	3	0.0085
stress_today	1.8	6	5	0.0060

runner_up tabpfn - Spearman rho = -0.03

